A Literature-Implied Generalized James-Space Subcase of a Bidual-Separability Problem

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Source and Original Problem

The source paper is Baudier–Lancien–Motakis–Schlumprecht, *Coarse and Lipschitz universality*, arXiv:2004.04806 [1]. Problem 5.15 asks whether the conclusion in their Theorem F can be sharpened from nonseparability of some even iterated dual of the target to nonseparability of the bidual itself.

Problem 5.15. *Assume that X is c_0, or any separable Banach space with non separable bidual X^{**} and $\ell_1 \not\subset X$ such that no spreading model generated by a normalized weakly null sequence is equivalent to the ℓ_1-unit vector basis. If X coarsely embeds into a Banach space Y, does it imply that Y^{**} is non separable?*

References

[Aha74] I. Aharoni, *Every separable metric space is Lipschitz equivalent to a subset of c_0^+*, Israel J. Math. **19** (1974), 284–291.

[AJO05] D. Alspach, R. Judd, and E. Odell, *The Szlenk index and local l_1-indices*, Positivity **9** (2005), no. 1, 1–44, DOI 10.1007/s11117-002-9781-0.

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[AD07] S. A. Argyros and P. Dodos, *Genericity and amalgamation of classes of Banach spaces*, Adv. Math. **209** (2007), 666–748.

[Ban32] S. Banach, *Théorie des opérations linéaires*, Warszawa, 1932.

The relevant subcase considered here is the case where the embedded space is $X = c_0$ and the target space is a generalized James space $\mathrm{J}(e_i)$ over an unconditional basis. The general problem is not claimed to be solved.

Supporting Literature

The supporting paper is Jackson–Krause–Sari, *On coarse geometry of separable dual Banach spaces*, arXiv:2501.01561 [2]. Their Theorem 29 says that for every unconditional basis (e_i),

$$c_0 \text{ coarsely embeds into } \mathrm{J}(e_i) \quad \Longleftrightarrow \quad c_0 \text{ linearly embeds into } \mathrm{J}(e_i).$$

We will show next that ℓ_1 assumption in (i) of the above Corollary is not necessary.

Theorem 29. Let (e_i) be an unconditional basis for a Banach space E, and $\mathrm{J}(e_i)$ be the James space over (e_i). Then c_0 coarsely embeds into $\mathrm{J}(e_i)$ if and only if c_0 linearly embeds into $\mathrm{J}(e_i)$.

Proof. Suppose c_0 does not linearly embed into $\mathrm{J}(e_i)$. Then by [BHO] the basis (u_i) of $\mathrm{J}(e_i)$ is boundedly complete. Suppose $\phi : c_0 \to \mathrm{J}(e_i)$ is a coarse embedding. The proof is a slight variation of the proof of Theorem 21 so we will only briefly indicate the required additional argument which additionally exploits the fact that the summing functional S is bounded on $\mathrm{J}(e_i)$ and the block sequences (w_i) with $S(w_i) = 0$ are unconditional (See Proposition 2.1

Idea of the Proof

The later theorem converts the nonlinear hypothesis in the source problem, "c_0 coarsely embeds into Y", into the linear conclusion "c_0 embeds linearly into Y" for the particular targets $Y = \mathrm{J}(e_i)$. Once a linear copy of c_0 is present, the bidual obstruction is standard: the second adjoint of an isomorphic embedding of c_0 into Y embeds $c_0^{**} = \ell_\infty$ into Y^{**}. Since ℓ_∞ is nonseparable, Y^{**} must be nonseparable.

Theorem

Theorem 1. *Let (e_i) be an unconditional basis for a Banach space E, and let $Y = \mathrm{J}(e_i)$ be the generalized James space over (e_i) in the sense of [2]. If c_0 coarsely embeds into Y, then Y^{**} is nonseparable.*

Proof. Assume that c_0 coarsely embeds into $Y = \mathrm{J}(e_i)$. By Theorem 29 of [2], c_0 linearly embeds into Y. Thus there is a bounded linear operator $T : c_0 \to Y$ and a constant $a > 0$ such that

$$\|Tx\| \geq a\|x\| \qquad (x \in c_0).$$

The second adjoint $T^{**} : c_0^{**} \to Y^{**}$ is also an isomorphic embedding. For completeness, let $S : T(c_0) \to c_0$ be the inverse of T on its range, so $\|S\| \leq a^{-1}$. If $\varphi \in B_{c_0^*}$, then φS is a bounded functional on the closed subspace $T(c_0)$ of Y, and Hahn–Banach extends it to some $\psi \in Y^*$ with $\|\psi\| \leq \|S\|$ and $T^*\psi = \varphi$. Hence

$$B_{c_0^*} \subset \|S\|\, T^*(B_{Y^*}).$$

For $x^{**} \in c_0^{**}$, this gives

$$\|x^{**}\| = \sup_{\varphi \in B_{c_0^*}} |x^{**}(\varphi)| \leq \|S\| \sup_{\psi \in B_{Y^*}} |x^{**}(T^*\psi)| = \|S\|\, \|T^{**}x^{**}\|.$$

Thus T^{**} is bounded below.

Since c_0^{**} is isometric to ℓ_∞, this embeds ℓ_∞ into Y^{**}. Therefore Y^{**} is nonseparable. $\square$

Scope and Limitations

This is a literature-implied partial subcase of Problem 5.15, not a full answer. It proves the bidual conclusion for targets of the form $Y = \mathrm{J}(e_i)$ and embedded source $X = c_0$. It does not address arbitrary Banach targets Y, nor the other source spaces allowed in Problem 5.15.

The supporting paper does not appear to state that it is answering Problem 5.15. The match is by direct specialization of its Theorem 29 plus the standard second-adjoint observation above. The run indexes were searched for arXiv:2004.04806, arXiv:2501.01561, generalized James-space keywords, “c_0 coarsely embeds”, and the “Y^{**} nonseparable” phrasing. A bounded web search on June 14, 2026 for exact phrases from Problem 5.15 found only the source paper.

Human Review Recommendation

Check that the notation $\mathrm{J}(e_i)$ and the unconditional-basis hypothesis in [2] cover exactly the generalized James targets intended here. Also verify the second-adjoint embedding step, since it is the only extra identification beyond the cited literature theorem.

References

[1] F. Baudier, G. Lancien, P. Motakis, and T. Schlumprecht, *Coarse and Lipschitz universality*, Fundamenta Mathematicae 254 (2021), no. 2, 181–214; arXiv:2004.04806.

[2] S. Jackson, C. Krause, and B. Sari, *On coarse geometry of separable dual Banach spaces*, arXiv:2501.01561, 2025.